

Equation of a circle:

$$(x - a)^2 + (y - b)^2 = r^2$$

Radius = r

Centre = (a, b)

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{Radius} = \sqrt{g^2 + f^2 - c}$$

Centre = $(-g, -f)$

Intersection of a circle and a straight-line:

$$(x - a)^2 + (y - b)^2 = r^2$$

$$y = mx + c$$

Substitute y in the circle equation with $mx + c$ and solve the quadratic in terms of x

Since a quadratic equation is formed, the discriminant can also be used to determine the number of points of intersection

$$b^2 - 4ac > 0 \rightarrow 2 \text{ points, the line is a chord}$$

$$b^2 - 4ac = 0 \rightarrow 1 \text{ point, the line is a tangent}$$

$$b^2 - 4ac < 0 \rightarrow 0 \text{ points, the line does not intersect the circle}$$

Tangents:

The tangent to a circle intersects it once, on its circumference.

The gradient of the tangent is perpendicular to the gradient of the centre to the point of intersection

The equation of a tangent can be found from its gradient and point of intersection

Intersection of two circles:

Two circles may intersect twice, once or never.

With two centres (x_1, y_1) and (x_2, y_2) , the distance d is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$r_2 - r_1 < d < r_2 + r_1 \rightarrow 2$ points of intersection

$r_2 - r_1 = d = r_2 + r_1 \rightarrow 1$ point of intersection

$d < r_2 - r_1, d > r_2 + r_1 \rightarrow 0$ points of intersection

To find the points of intersection:

- 1) Re-arrange both equations so that RHS=0
- 2) Equate the equations to each other (questions will be framed such that x^2 and y^2 cancel)
- 3) You will be left with a line on which the point(s) of intersection lie
- 4) Substitute the line into one of the equations to get the x or y coordinate
- 5) Substitute the coordinate into one of the equations to get the other coordinate